# The AI-MDM Pipeline Works, but Creates a Governance Black Box

## Metadata Catalog Adds Missing Traceability, Ownership, and Quality Gates

### What AI Delivers

- **Auto-extraction of unit fields from NPD specifications**
  - AI Layer 1: VLM + LLM extract 15 fields per product (master, inner, base units)
  - Replaces manual R&D data entry
- **Elimination of manual R&D data entry workload**
  - Saves data entry lead-time per product
- **Faster time-to-market for new products**
  - Reduces NPD cycle time
  - Enables rapid product launches
  - Accelerates NPD-to-ERP distribution

### THE GAP: 4 UNANSWERED QUESTIONS

| Gap | Implication |
|---|---|
| **Where did this ERP value come from?** | Cannot trace field value back to source document |
| **Which fields were AI-generated?** | No transformation transparency |
| **Who's responsible for this wrong value?** | No accountability routing |
| **Which field should I use for analytics?** | No canonical vs derived clarity |

### METADATA SOLUTION

| Function | Enables |
|---|---|
| **Data Lineage** | Data Source traceability |
| **Data Classification** | Data Transformation tags |
| **Data Ownership** | DAMA roles at every asset |
| **Schema Design** | Canonical vs derived data clarity |

# Why a Data Catalog? Making Pipeline Knowledge Persistent

## OpenMetadata Capabilities for MDM Governance

**KNOWLEDGE UNAVAILABLE WHEN NEEDED MOST**

- Pipeline builders know **HOW** transformation logic in YAML/code
- Data stewards know **WHICH** approved records in email/spreadsheets
- Department leads know **WHO/WHAT** ownership and definitions tribal)

**Knowledge unavailable when issues arise or new members join**

Data Catalog Platform

| | Capability | What It Governs | Governance Value |
|---|---|---|---|
| 1 | **Data catalog** | Registers and describes all data assets (schemas, tables, columns, pipelines) | Centralized asset inventorysingle source of truth for "what exists" |
| 2 | **Classifications and tags** | Labels every asset and column with governance metadata (MDM role, transformation type, sector) | Transformation transparency VLM/LLM/RuleBased tags show HOW field was produced |
| 3 | **Table/Field-level lineage** | Maps how every value flows from source document to ERP field | Source traceability |
| 4 | **Business glossary** | Defines canonical terms linked to catalog columns | Shared vocabulary data definitions defined once, linked to all instances |
| 5 | **Ownership model** | Assigns data owner, steward, and custodian to every asset | Accountability routing, R&D owns canonical, Sales owns sales_view |
| 6 | **Data product** | Packages canonical layer as consumer contract with SLA and quality terms | Governance by contract, expectations explicit, not tribal |

# 1) Schema Boundaries Enforce Accountability — One Team, One Schema

Database: Product_Master_Data | Topic: Unit of Measurement | Active: Poultry

**Database: product_master_data**

**Schema (Container)**

**source_documents**
- NPD specification PDFs - AI extraction origin

**canonical**
- Single source of truth - all sectors combined

**canonical_view**
- Sector-filtered views ownership by R&D

**production_view**
- Manufacturing units

**sales_view**
- Customer-facing units
- **Dependency enforced**

**Owner: MDM Platform (Custodian)**

**reference_data**
- Lookup tables - ERP mappings & RAG samples

**Tables (asset)**

npd_spec_poultry
- PDF
- Excels

Owner: Poultry R&D

npd_spec_swine

Owner: Poultry R&D

npd_spec_processed

Owner: Poultry R&D

unit_of_measure
- 17 data fields
- Consolidate all sectors

Owner: MDM Council

uom_poultry
- 17 data fields (filtered)
- *WHERE sector='Poultry'*

Owner: Poultry R&D

uom_swine

Owner: Swine R&D

uom_processed

Owner: Processed R&D

prod_uom_poultry
- 2+ fields
- Rulebased transformation

Owner: Prod Planning Poultry

prod_uom_swine

Owner: Prod Planning Swine

prod_uom_processed

Owner: Prod Planning Processed

sales_uom_poultry
- 2+ fields
- Rulebased transformation

Owner: Sales Op Poultry

sales_uom_swine

Owner: Sales Op Swine

sales_uom_processed

Owner: Sales Op Processed

Unit Mappings

RAG pool

**Reasoning**

**WHY:** Different NPD templates per sector.
**OWNER:** Sector R&D teams (Each sector R&D own source accuracy)
**AUDIT:** Regulatory proof of data origin (document hash → extraction log → canonical)
**TECH:** VLM extraction configs vary by sectors.

- **WHY:** Single source of truth (one place to fix canonical field definitions)
- **OWNER:** MDM Stewardship Council (cross-sector policy authority)
- **ANALYTICS:** Cross-sector analysis enabled.
- **SCALE:** Add sector = add rows (not rebuild table)

- **WHY:** Sector ownership isolation (Poultry R&D accountable for Poultry data only)
- **ACCESS:** Sector R&D sees only their sector data.
- **GOVERNANCE:** Council decides "what's valid", R&D decides "is this record valid?"
- **SCALE:** Add sector = 1 SQL view filter (WHERE sector='fish'), not table rebuild

- **WHY:** Manufacturing context (translates canonical → plant floor terminology)
- **OWNER:** Production Planning (defines what manufacturing operations need)
- **DECOUPLE:** Production logic changes isolated from Sales/SCM

- **WHY:** Customer-facing context (ERP/POS units for order processing)
- **DEPENDENCY:** Sales cannot run until Production says "my data is correct"
- **CONSISTENCY:** Uses Production's approved data (not raw canonical)
- **AUDIT:** Sales data issue → trace Production → Canonical → Source (full chain)

- **WHY:** Change business rules by editing table, not rewriting code
- **UNIVERSAL:** Unit conversions shared.
- **SECTOR-SPECIFIC:** ERP codes vary (Poultry 'CASE' vs Swine 'CS' via sector column)
- **OWNER:** MDM Platform (Data Custodian maintains lookup tables)

# 2) Four Tag Dimensions Work Together to Enable Root Cause Diagnosis

Database: Product_Master_Data | Applied: Table & Column Levels

## Classification Hierarchy: Governance Encoded as Tags

### MDM Classification

**Applied to:** All tables
**Purpose:** Role in MDM lifecycle

| Tag | Description |
|---|---|
| **Canonical** | Single source of truth, all dept views derive from this |
| **Derived** | Department-specific view via data transformation |
| **Source** | Raw NPD specification documents (extraction origin) |
| **Configuration** | YAML rules, VLM templates, transformation configs |
| **Dependency Enforced** | Blocked view until upstream approved |

### Sector

- **Applied to:** All tables
- **Purpose:** Business unit ownership

| Tag | Description |
|---|---|
| **Poultry** | Poultry business sector |
| **Swine** | Swine business sector |
| **Processed_Food** | Ready-to-eat products |

### Data Type

- **Applied to:** All tables
- **Purpose:** Functional classification

| Tag | Description |
|---|---|
| **MasterData** | Product canonical + dept views |
| **ReferenceData:** | Lookup tables (unit conversions, unit masters) |
| **ConfigurationData** | YAML storage, transformation rules |
| **MappingData** | ERP routing, field cross-references |
| **MetadataRegistry** | Extraction logs, audit trails |

### Transform Classification

- **Applied to:** All Individual columns
- **Purpose:** How field value was produced

| Tag | Description |
|---|---|
| **VLM** | Extracted document via VLM |
| **LLM** | Data generated by GenAI |
| **RuleBased** | Conditional YAML logic |
| **Lookup** | Reference table mapping |
| **DirectCopy** | Pass-through unchanged |
| **Hardcoded** | Constant value from config |

## Reasoning

- **WHY:** Identifies lifecycle stage (source → canonical → derived)
- **USE:** Filter canonical vs derived tables
- **GOVERN:** Know dependencies before making changes

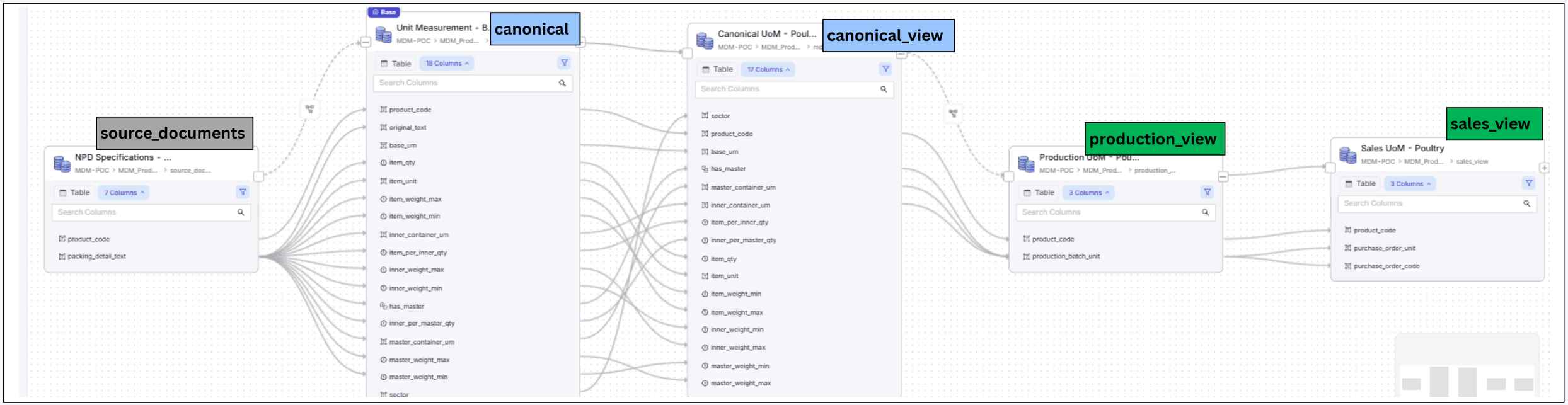
- **WHY:** Routes data quality issues to correct business unit
- **USE:** Access control (Poultry R&D sees only Poultry data)
- **SCALE:** Add sector = add tags

- **WHY:** Separates master data from supporting lookups.
- **USE:** Platform owns Reference, Business owns Master
- **CUSTODIAN:** Clear responsibility boundaries

- **WHY:**
  - Documents field production method
  - Different production methods require different quality controls.
- **USE:**
  - Identify root cause location
  - Determine validation approach
  - Assess change impact

# 3) Field-Level Lineage Enables Root Cause in Minutes, Not Days

## Complete Traceability from Source to Consumption




### The MDM Challenge:

- One canonical field flows to 10+ downstream systems
- Each system transforms it differently (batch units, order units, invoice units)
- **Table lineage** says "production view uses canonical" (not helpful)
- **Field lineage** shows a production field derives from few canonical fields via RuleBased logic" (actionable)

### 4 CRITICAL CAPABILITIES ENABLED WITH FIELD-LEVEL LINEAGE

| Capabilities | Description |
|---|---|
| **Root Cause Analysis** | • **Trace backward:** Invoice error → Sales field → Production field → Canonical field → Source extraction<br>• **Identify:** Exact transformation that introduced error (AI extraction, business rule, lookup)<br>• **Result:** Fix at source, automatic propagation (not symptom patching) |
| **Impact Assessment** | • **Trace forward:** Canonical field → Production (batch_unit) → Sales (order_unit) → Finance, Logistics, Inventory<br>• **Identify:** Full downstream impact before making changes<br>• **Result:** Prevent cascade failures across departments |
| **Regulatory Compliance** | • **Prove:** Field-by-field chain of custody from source document to final report<br>• **Show:** Transformation method, approval status, responsible party at each stage<br>• **Result**: Audit-ready in minutes (regulators require field-level proof) |
| **Quality Monitoring** | • **Tag:** Each field with transformation method (AI, RuleBased, Lookup)<br>• **Monitor:** Error rates by transformation type (VLM, LLM reasoning, rule logic)<br>• **Target:** Improvements where needed (retrain AI, refine rules) |

# 4) Business Glossary: Shared Language Across Stakeholders

Database: Product_Master_Data | Two Glossaries: Technical Governance + Business UoM

## WHY GLOSSARIES MATTER:

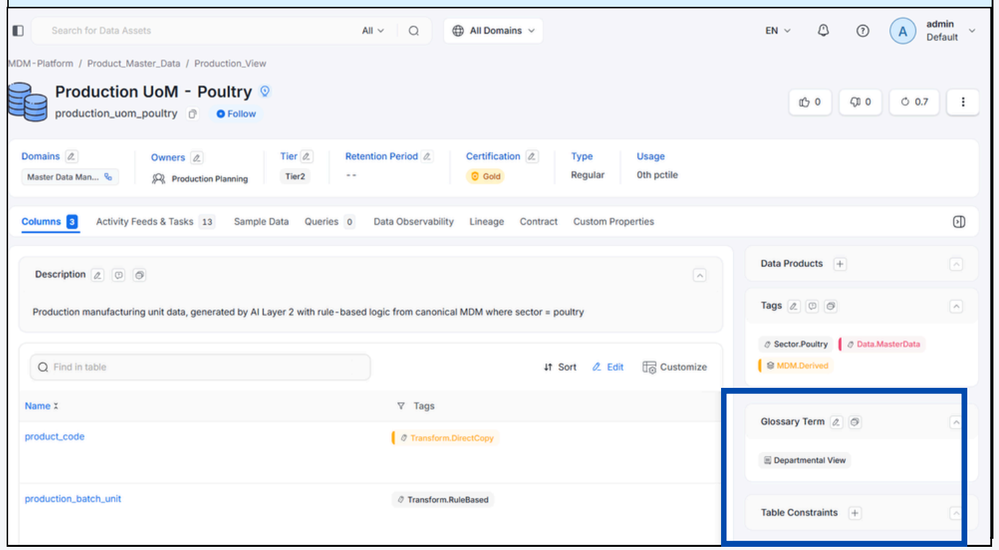
**Terms link to assets:**

- **Prevent miscommunication:** standardize vocabulary across teams
- **Traceability:** Link business terms to technical assets
- **Enable self-service:** users understand data without asking IT
- **Support onboarding:** new team members learn domain language

**Examples**

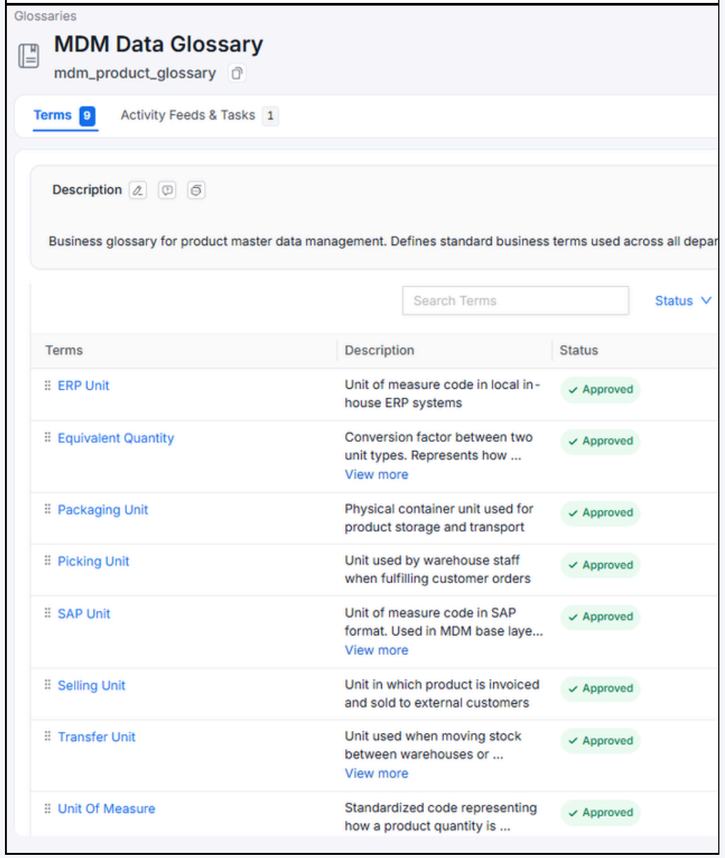
- "Canonical Data" → Links to canonical schema tables
- "Packaging Unit" → Links to master_container_um column
- "Departmental View" → Links to production_view, sales_view

**Result:** Shared understanding without technical jargon overload



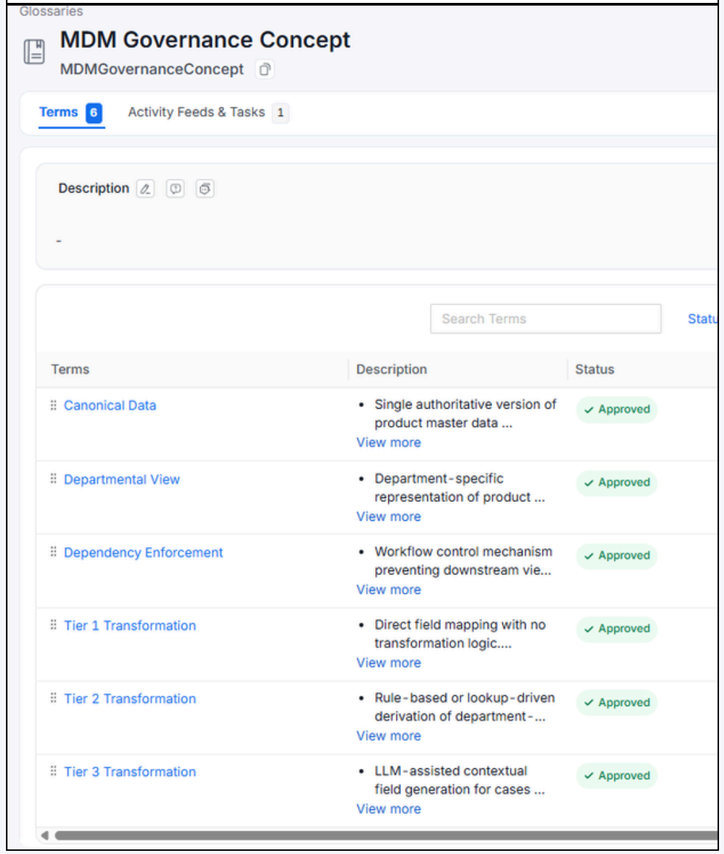

## Glossary 1: MDM Data Glossary

- **PURPOSE:** Define business terminology
- **AUDIENCE:** Operations, R&D, Finance.
- **EXPLAINS:** How catalog works (governance workflows)




## Glossary 2: MDM Governance Concept

- **PURPOSE:** Define WHAT business terms mean
- **AUDIENCE:** Platform teams, data stewards
- **EXPLAINS:** WHAT units mean in business context (departmental usage)



# 5) Ownership & Accountability: DAMA Roles Mapped to Catalog Structure

Database: Product_Master_Data | Clear Responsibility Chains

| Domain | Team | DAMA Role | Schema/Assets Owned | Accountability |
|---|---|---|---|---|
| Master Data Management | MDM Platform | **Data Custodian** | • *reference_data*<br>• *source_documents* | • Maintain transform logic<br>• Operate data pipelines<br>• Mangement Data Catalog<br>• Owns Reference Data Table<br>• Troubleshoot technical issues |
| | MDM Stewardship Council | **Data Owner** | • *canonical* | • Sets canonical data policy<br>• Cross-sector governance<br>• Approves schema changes<br>• Defines data standards |
| Product Management | Research & Development | **Data Owner + Data Steward** | • *canonical_views*<br>• *source_documents* | • Review AI extractions for Poultry products<br>• Validate NPD specification quality /Approve canonical records<br>• Accountable for its respective sector data accuracy |
| Manufacturing | Production Planning | **Data Steward** | • *production_view* | • Define manufacturing requirements<br>• Validates production view transformation logic<br>• Approves transformed production data |
| | | **Data Consumer** | • *Uses: production_uom tables* | • Use data for manufacturing planning. |
| Sales | Sales Operation | **Data Steward** | • *sales_view* | • Validates sales view transformation logic<br>• Approves transformed sales data |
| | | **Data Consumer** | • *Uses: sales_uom tables* | • Use data for order processing and sales analytics |

## DATA GOVERNANCE PRINCIPLES TERMS

**Domain**
- A business area or functional category that groups related data assets.

**Team**
- A group of people responsible for governing and maintaining data assets within a domain
- Teams are assigned ownership of domains and the assets within them.

**Data Owner**
- Accountable for data quality, sets policy, approves changes

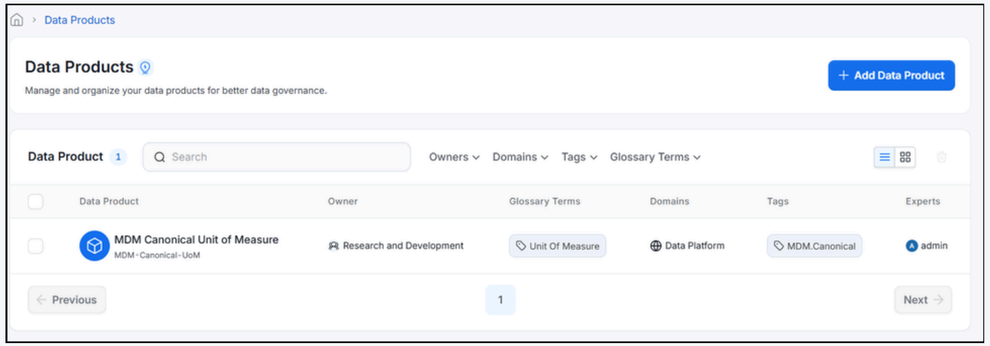
**Data Steward**
- Executes quality controls, reviews data, approves records

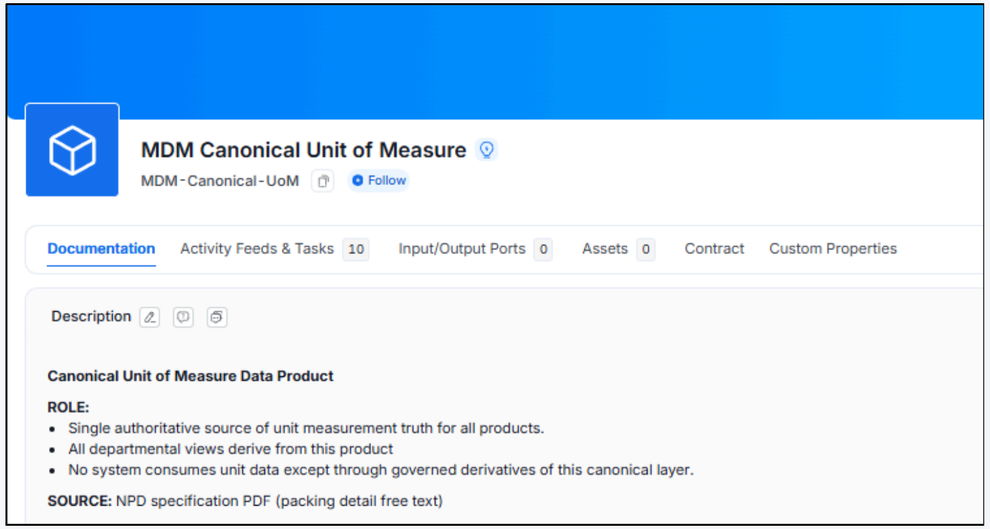
**Data Custodian**
- Maintains technical infrastructure, operates pipelines, manages catalog

**Data Consumer**
- Uses data for business operations
- Reports quality issues

# 6) Data Product: Governance by Contract, Not Tribal Knowledge

## MDM-Canonical-UoM — Self-Service Unit Data for Analytics Consumers




**Packages canonical master data as governed, analytics-ready offering:**

- **Certified data source:** trust without validation
- **Clear contract:** SLA, refresh, quality guarantees
- **Self-service access:** use directly, no IT requests
- **Defined ownership:** know who to contact

**Use for Data Analytics such as:**

- Revenue reporting (unit price × quantity sold)
- Batch planning (production units → packaging units)
- Inventory valuation (cost per base unit)
- Regulatory compliance (labeling unit accuracy)

**WITHOUT DATA PRODUCT**

- Table exists in catalog
- Metadata describes fields
- **But:** readiness unknown, no SLA, unclear who validates
- Result: consumers ask people, not systems

**WITH DATA PRODUCT:**

- Canonical layer packaged as contract
- Ownership declared (who guarantees quality)
- Service terms explicit (when it's ready, how to access)
- Result: self-service discovery, clear accountability

Metadata answers 'what is this?' — **Data Product answers 'can I trust this, and what are the rules?**

# Applied to the unit measurement data topic — each metadata element closes a specific diagnostic gap identified in the main analysis

| DAMA Area | Diagnostic Finding | Metadata Element | How it mitigate issue |
|---|---|---|---|
| **Document & Content Management** | PDFs invisible to MDM — unit data trapped in unstructured specs | • **Schema Design:** Source_Documents schema registers NPD PDFs as governed assets<br>• **Lineage:** Field-level links trace all 15 canonical fields back to source PDF | Every unit field traces to a specific source document |
| **Data Model & Design** | MDM is product registry — unit measurement fields not in model | • **Schema Design:** Canonical base table established as single source of truth<br>• **Classification:** MDM.Canonical vs MDM.Derived tags distinguish source from derivatives | One authoritative unit model departmental views are tagged derivatives |
| **Master and Reference Data** | Unit code inconsistency across department systems — same SAP code interpreted differently | • **Schema Design:** Reference_Data schema with sector column separates business unit codes<br>• **Ownership:** MDM Platform (Custodian) owns reference tables for centralized governance | Unit codes consistent across all ERP systems from one governed translation source |
| **Data Quality Management** | No validation rules non-specialist entry risk | • **Classification:** Transform tags (VLM, LLM, RuleBased) identify which fields need validation<br>• **Ownership:** R&D Stewards review and approve AI-extracted canonical fields<br>• **Workflow:** Dependency enforcement blocks Sales until Production approved | AI-extracted fields require human validation |
| **Metadata Management** | No lineage, no field ownership, tribal knowledge — every issue untraceable | • **Lineage:** 4-stage chain (NPD → Canonical → Production → Sales) with 20+ field mappings<br>• **Classification:** Transform tags document how each field value was produced<br>• **Schema Design:** Schema-as-boundary ensures each schema has one accountable team | Root cause of wrong unit field traceable in shorter time |
| **Data Governance** | Policies exist but not operationalized | • **Ownership:** DAMA roles (Owner, Steward, Custodian) declared at every asset<br>• **Schema Design:** Schema = ownership boundary (one team per schema)<br>• **Workflow:** Dependency tags enforce workflow rules in catalog metadata | Governance rules enforceable by reference, not just paper documentation |
| **Quantifiable Impact** | • **Time Savings -** Efficient Root cause diagnosis<br>• **Data Quality Improvement -** Data Validation<br>• **Governance Maturity -** LV3 Defined → LV4 Managed | | |

# Governance in Practice: How All Elements Work Together

## Issue Resolution Using Complete Catalog

**ISSUE:** production_batch_unit = "BAG" (should be "CASE") for PO-12345
**Impact:** Wrong manufacturing quantities scheduled

### DATA CONSUMER

**1 Detech Issue**
- Create Data Quality Issue Ticket

**5 Receive Fix**
- Updated Data
- Close Ticket

### DATA CUSTODIAN

**2 Triage and Routing (~10 minutes)**

| Catalog Element | Discovery |
|---|---|
| 1) Schema | Issue in Production_View schema<br>Check if transform logic or upstream data |
| 2) Classification | Problematic field has Transform.RuleBased tag so rule logic is correct, **input field must be wrong** |
| 3) Lineage | Trace production_batch_unit back to **master_container_um** (LLM extraction) |
| 5) Ownership | canonical_uom_poultry owned by Poultry R&D |

**Diagnosis:** LLM misread "CASE" as "BAG" in NPD PDF
**Fix Location:** *master_container_um* at Canonical
**Assign:** Poultry R&D (Data Steward)

**4 Automated Regenerate Downstream**
- Production regenerates after approval, then Sales unblocks

### DATA STEWARD

**3 Fix at source (~30 min)**

| |
|---|
| Review NPD specification PDF for PO-12345 |
| Confirm original: Master = CASE |
| Correct canonical (~~BAG~~ → CASE) |
| Mark record: "Approved" |

**Key Takeaway**

| Without Data Catalog | With Data Catalog |
|---|---|
| 1. Manual code reading | 1. Tag lookup (9 min) |
| 2. Email escalation (3-5 teams) | 2. Auto-route to Poultry R&D |
| 3. Fix symptom (recurs) | 3. Fix root cause (permanent) |
| 4. ~2-3 days resolution | 4. ~1-2 hour resolution |